

Randomised Experiments with Clusters

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- ▶ These groups of individuals are *clusters*
- ▶ (Let's start with easy case: all clusters same size ...)
(Gerber and Green 2012)

Variance

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$$SE(\widehat{ATE}) = \sqrt{\frac{1}{k-1} \left[\frac{m \text{Var}(\overline{Y}_j(0))}{N-m} + \frac{(N-m) \text{Var}(\overline{Y}_j(1))}{m} + 2 \text{Cov}(\overline{Y}_j(0), \overline{Y}_j(1)) \right]}$$

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- ▶ If $\text{Var}(\overline{Y}_j(1))$, $\text{Var}(\overline{Y}_j(0))$ are small, SE is small

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Middleton and Aronow (2015)

Bias

Alternative estimator 1: Difference in totals (not means)

$$\widehat{ATE} = \frac{k_T + k_C}{N} \left(\frac{\sum Y_i(1)|T_i = 1}{k_T} - \frac{\sum Y_i(0)|T_i = 0}{k_C} \right)$$

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- ▶ *Unbiased* for true difference in *totals*
- ▶ But, doesn't include cluster sizes, so may be high variance
- ▶ I.e., more students per classroom doesn't help precision here (unlike above)

Alternative estimator 2: Horvitz-Thompson estimators

$$\hat{\tau}^{HT} = \frac{1}{N} \sum_{i:T_i=1} \frac{Y_i}{\pi_i} - \frac{1}{N} \sum_{i:T_i=0} \frac{Y_i}{1 - \pi_i}$$

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 - ▶ $\pi_i = E\left(\frac{n_j}{n}\right)$

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- ▶ Unbiased for difference
- ▶ `estimatr::horvitz_thompson()`

An Example

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- ▶ Measure ideology, which varies by city and voter
- ▶ Do 2 hypothetical assignments
 - ▶ Individual-level assignment to news
 - ▶ City-level cluster assignment to news
- ▶ Use RI to find SE in each case

An Example

```
library(fabricatr)
library(randomizr)
library(tidyverse)

# Make data and visualise

set.seed(95852894)

voters <- fabricate(
  N = 1000,
  city_id = rep(1:10, 100),
  ideology = draw_normal_icc(mean = 0, N = N,
                             clusters = city_id, ICC = 0.7)
  city_id_fac = as.factor(city_id),
  city_id_fac = fct_reorder(city_id_fac, ideology)
)
```

An Example

```
table(voters$city_id)
```

1	2	3	4	5	6	7	8	9	10
100	100	100	100	100	100	100	100	100	100

An Example

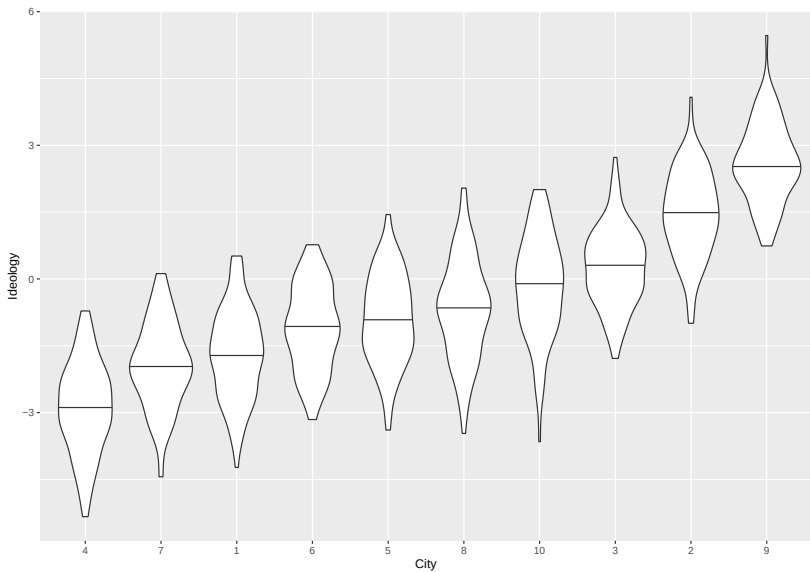
```
table(voters$city_id)
```

```
 1    2    3    4    5    6    7    8    9   10  
100 100 100 100 100 100 100 100 100 100
```

```
voters |> group_by(city_id) |>  
  summarise(mean_ideo = mean(ideology)) |>  
  arrange(mean_ideo)
```

```
# A tibble: 10 x 2  
  city_id mean_ideo  
    <int>     <dbl>  
1       4    -2.90  
2       7    -1.94  
3       1    -1.68  
4       6    -1.10  
5       5    -0.903  
6       8    -0.686  
7      10    -0.161  
8       3     0.267  
9       2     1.46  
10      9     2.60
```

An Example



An Example

Assignments

```
voters <- voters |>
  mutate(
    tr_ind = sample(rep(0:1, nrow(voters) / 2)),
    tr_cl = cluster_ra(clusters = voters$city_id, m = 5))
```

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```

```
voters |> count(tr_ind)
```

	tr_ind	n
1	0	500
2	1	500

```
voters |> count(tr_cl)
```

	tr_cl	n
1	0	500
2	1	500

An Example

Conditions by city, individual-level assignment:

```
table(voters$city_id, voters$tr_ind)
```

	0	1
1	48	52
2	47	53
3	52	48
4	47	53
5	51	49
6	49	51
7	49	51
8	53	47
9	48	52
10	56	44

An Example

Conditions by city, cluster-level assignment:

```
table(voters$city_id, voters$tr_cl)
```

	0	1
1	0	100
2	100	0
3	100	0
4	0	100
5	100	0
6	0	100
7	0	100
8	0	100
9	100	0
10	100	0

An Example

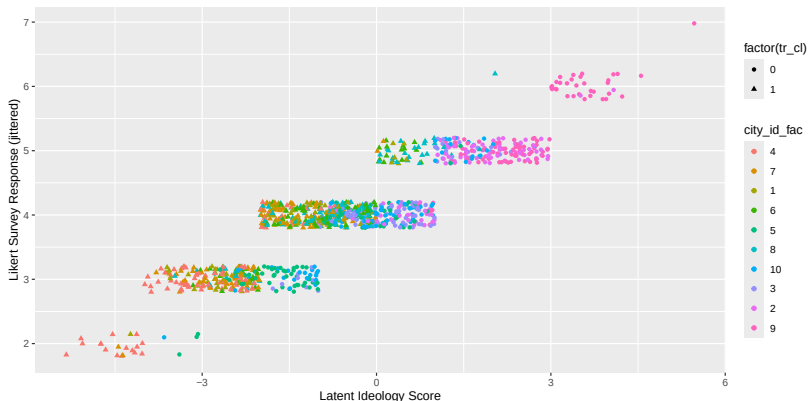
Draw responses under each assignment:

```
voters <- voters |>
  mutate(
    response_ind = draw_likert(x = ideology + tr_ind, m
    response_cl = draw_likert(x = ideology + tr_cl, m
```

An Example

Draw responses under each assignment:

```
voters <- voters |>
  mutate(
    response_ind = draw_likert(x = ideology + tr_ind, m = 10),
    response_cl = draw_likert(x = ideology + tr_cl, m = 10)
```



An Example

Analyse precision with RI:

- ▶ Do 1000 individual-level hypothetical assignments

An Example

Analyse precision with RI:

- ▶ Do 1000 individual-level hypothetical assignments
- ▶ Do 1000 cluster-level hypothetical assignments

An Example

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- ▶ Use sharp null $H_0 : \tau_i = 0$ to impute unobserved potential outcomes

An Example

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- ▶ Do 1000 individual-level hypothetical assignments
- ▶ Do 1000 cluster-level hypothetical assignments
- ▶ Use sharp null $H_0 : \tau_i = 0$ to impute unobserved potential outcomes
- ▶ Calculate $\widehat{ATE}$ in each case

An Example

Analyse precision with RI:

- ▶ Do 1000 individual-level hypothetical assignments
- ▶ Do 1000 cluster-level hypothetical assignments
- ▶ Use sharp null $H_0 : \tau_i = 0$ to impute unobserved potential outcomes
- ▶ Calculate $\widehat{ATE}$ in each case
- ▶ Compare distribution of $\widehat{ATE}_{\text{Individual}}$ to distribution of $\widehat{ATE}_{\text{Cluster}}$

An Example

Analyse precision with RI:

- ▶ Do 1000 individual-level hypothetical assignments
- ▶ Do 1000 cluster-level hypothetical assignments
- ▶ Use sharp null $H_0 : \tau_i = 0$ to impute unobserved potential outcomes
- ▶ Calculate $\widehat{ATE}$ in each case
- ▶ Compare distribution of $\widehat{ATE}_{\text{Individual}}$ to distribution of $\widehat{ATE}_{\text{Cluster}}$
- ▶ With partner, sketch the two randomisation distributions!

An Example

Analyse precision with RI:

```
n_sims <- 1000
df_ates <- tibble(ate_ind = NA,
                  ate_cl = NA)

for(idx in 1:n_sims){

  voters <- voters |>
    mutate(
      hyp_tr_individ = sample(rep(0:1, nrow(voters) / 2)),
      hyp_tr_cl = cluster_ra(clusters = voters$city_id, m = 5))

  ate_ind <- mean(voters$response_ind[voters$hyp_tr_ind == 1]) -
    mean(voters$response_ind[voters$hyp_tr_ind == 0])

  ate_cl <- mean(voters$response_cl[voters$hyp_tr_cl == 1]) -
    mean(voters$response_cl[voters$hyp_tr_cl == 0])

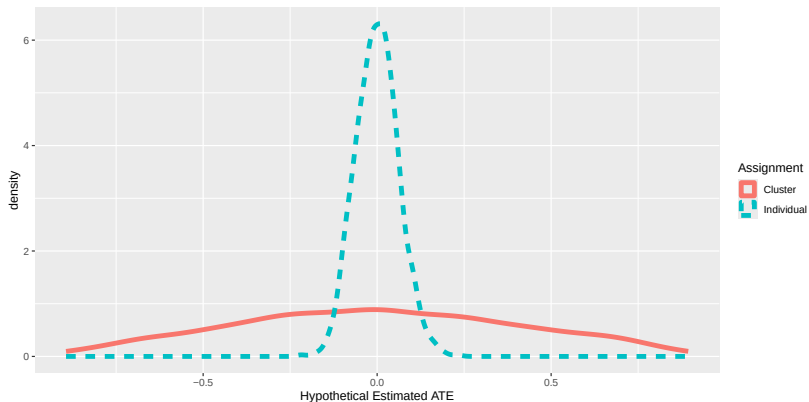
  df_ates[idx, "ate_ind"] <- ate_ind
  df_ates[idx, "ate_cl"] <- ate_cl
}
```

An Example

Analyse precision with RI:

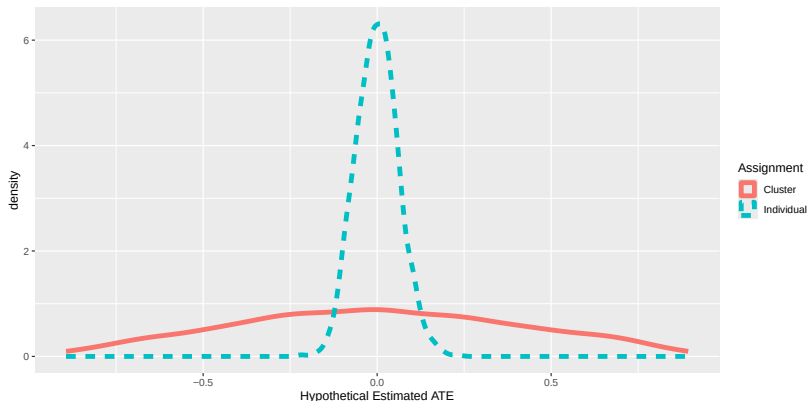
An Example

Analyse precision with RI:



An Example

Analyse precision with RI:



```
# A tibble: 1 x 2  
  se_ind se_cl  
  <dbl> <dbl>  
1 0.0614 0.408
```

An Example

Same as “cluster-robust SEs” from regression:

```
# A tibble: 1 x 2  
  se_ind se_cl  
  <dbl> <dbl>  
1 0.0614 0.408
```

An Example

Same as “cluster-robust SEs” from regression:

```
# A tibble: 1 x 2
```

```
  se_ind se_cl
```

```
    <dbl> <dbl>
```

```
1 0.0614 0.408
```

```
# Est regression using last hyp Tr assg (DiM biased!):
```

```
lm_out <- lm(response_cl ~ hyp_tr_cl, data = voters)
```

```
# "Cluster-robust" SEs (CR2's, also heterosced robust):
```

```
lm_rob_cl <- lm_robust(response_cl ~ hyp_tr_cl, data = voters)
```

An Example

Same as “cluster-robust SEs” from regression:

```
# A tibble: 1 x 2
```

```
  se_ind se_cl
```

```
  <dbl> <dbl>
```

```
1 0.0614 0.408
```

```
# Est regression using last hyp Tr assg (DiM biased!):
```

```
lm_out <- lm(response_cl ~ hyp_tr_cl, data = voters)
```

```
# "Cluster-robust" SEs (CR2's, also heterosced robust):
```

```
lm_rob_cl <- lm_robust(response_cl ~ hyp_tr_cl, data = voters)
```

```
# A tibble: 2 x 4
```

```
  model      term      estimate std.error
```

```
  <chr>    <chr>      <dbl>    <dbl>
```

```
1 LS SEs  hyp_tr_cl  -0.242    0.0525
```

```
2 CR2 SEs hyp_tr_cl  -0.242    0.418
```


Next:
Regression and Experiments
PS3 due today

References I

- Gerber, Alan S., and Donald P. Green. 2012. *Field Experiments: Design, Analysis, and Interpretation*. New York, NY: WW Norton.
- Middleton, J. A., and P. M. Aronow. 2015. “Unbiased Estimation of the Average Treatment Effect in Cluster-Randomized Experiments.” *Statistics, Politics, and Policy* 6 (1-2): 39–75.